



**Verified Carbon
Standard**

26MWP SOLAR PHOTOVOLTAIC POWER PLANT IN BAVET CITY, CAMBODIA

Document Prepared by Shell Eastern Trading (Pte) Ltd

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1 PROJECT DETAILS

1.1 Summary Description of the Project

26MWp Solar Photovoltaic Power Plant in Bavet City, Cambodia (hereafter referred to as “the project”) is a solar power project with total installed capacity of 26MWp, located in Monourom village, Bavet City, Svay Rieng Province, the Kingdom of Cambodia. The project is invested by Green Sustainable Ventures (Cambodia) Co., Ltd. (hereafter referred to as “the project owner”).

The purpose of the project is to utilize the solar energy at the project site to generate and supply electricity to Electricite Du Cambodge Grid (EDC) which is dominated by coal-fired power stations. Prior to the implementation of the project, the electricity delivered to the grid by the project activity would be generated by the operation of grid-connected power plants and by the addition of new generation sources into the EDC. This is the scenario that existed prior to the implementation of the project activity and is also the baseline scenario.

The project is a renewable energy project and is expected to supply an annual average of 37,085 MWh of zero-emission electricity to EDC during the whole project lifetime of 25 years, which replace the power generation of those fossil fuel-fired power plants delivered to the EDC under the baseline scenario.

The project started construction on 13-Apr-2021 and commissioned on 27-Jan-2022. The expected total emission reductions are 185,314 tCO₂e during the first crediting period from 27-Jan-2022 to 26-Jan-2029, with average annual emission reductions of 26,473 tCO₂e.

1.2 Sectoral Scope and Project Type

Sectoral Scope 1: Energy industries (renewable - / non-renewable sources)

The project is not a grouped project.

1.3 Project Eligibility

The scope of the VCS Program includes:

1) The six Kyoto Protocol greenhouse gases: The emissions reduction of the project comes from two sources: CO₂ emissions from the production of the equivalent amount of electricity replaced by the project that would otherwise have been purchased from the fossil fuel fired power plants of EDC. Thus, the project applicable to this scope.

2) Ozone-depleting substances: NA

3) Project activities supported by a methodology approved under the VCS Program through the methodology approval process: NA.

4) Project activities supported by a methodology approved under a VCS approved GHG program, unless explicitly excluded under the terms of Verra approval: The methodology ACM0002 (Version 21.0) of the project utilized is a methodology approved under CDM Program, that is a VCS approved GHG program.

5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements: NA.

Meanwhile, Cambodia is Least Developed Country as designated by the United Nations¹, the project does not belong to the projects excluded in Table 1 of VCS Standard 4.3.

Thus, the project is eligible under the scope of VCS program.

1.4 Project Design

- ☒ The project includes a single location or installation only
- ☐ The project includes multiple locations or project activity instances, but is not being developed as a grouped project
- ☐ The project is a grouped project

Eligibility Criteria

Not applicable as the project is not grouped project.

1.5 Project Proponent

Organization name	Green Sustainable Ventures (Cambodia) Co., Ltd.
Contact person	Charles Tan
Title	Director
Address	No.6 Street No. 588. Sangkat Boeung Kok 11, Khan Toul Kork, Phnom Penh, Cambodia.

¹ <https://www.un.org/en/conferences/least-developed-countries>

Telephone	+60-125228603
Email	charlestan@pestech-international.com

1.6 Other Entities Involved in the Project

Organization name	Shell Eastern Trading (Pte) Ltd.
Role in the project	VCUs buyer
Contact person	Jack Chong
Title	Originator
Address	The Metropolis Tower 1, 9 North Buona Vista Drive, #07-01, Singapore 138588
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Organization name	Climate Bridge (Shanghai) Ltd.
Role in the project	Consultancy
Contact person	Gao Zhiwen
Title	General Manager
Address	Block B, Level 24, Jiangong Mansion, 33 Fushan Road, Pudong New Area, Shanghai, China 200120
Telephone	+86-2162462036
Email	<u>gao.zhiwen@climatebridge.com</u> ; <u>projects@climatebridge.com</u>

1.7 Ownership

The project owner of the project is Green Sustainable Ventures (Cambodia) Co., Ltd. The approval of Environmental Impact Assessment (EIA), Project Approval, and the business license of the project owner is evidence for legislative right. Besides, the purchasing power agreement are the evidence for the ownership of the plant, equipment and power generating.

1.8 Project Start Date

27-Jan-2022 (operation start date is the date on which the project began generating electricity and the project activity began generating GHG emission reductions).

1.9 Project Crediting Period

The project adopts seven years, twice renewable crediting period for a total of 21 years, and the first crediting period is from 27-Jan-2022 to 26-Jan-2029.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	☒
Large project	☐

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
27-Jan-2022~31-Dec-2022	24,959
01-Jan-2023~31-Dec-2023	26,749
01-Jan-2024~31-Dec-2024	26,615
01-Jan-2025~31-Dec-2025	26,482
01-Jan-2026~31-Dec-2026	26,349
01-Jan-2027~31-Dec-2027	26,217
01-Jan-2028~31-Dec-2028	26,086
01-Jan-2029~26-Jan-2029	1,857
Total estimated ERs	185,314
Total number of crediting years	7

Average annual ERs

26,473

1.11 Description of the Project Activity

The project started construction on 13-Apr-2021 and commissioned on 27-Jan-2022. The total installed capacity is 26MWp which consists of 57,639 pieces of the solar PV cell modules with unit capacity of 455Wp. The conversion ratio of inverters from direct current (DC) to alternating current (AC) is 1.3, i.e. DC/AC=1.3. Thus, the total installed capacity is 20MWac² on the side of alternating current. The project lifetime of solar PV cell modules is 25 years. The main technical parameters of the key equipment of the project are illustrated in below table.

Table 1-1. The key technical parameters of solar cell modules and inverters³

Item		Unit	Project
Solar cell module	Type	-	Q. PEAK DUO XL-G9.3 455W
	Number	-	57,639
	Rated Power	Wp	445
	Rated Voltage	V	44.06
	Rated Current	A	10.10
	Open Circuit voltage	V	53.15
	Short Circuit current	A	10.62
	Efficiency	%	≥20
	Lifetime	year	25
	Manufacturer	-	Hanwha Q CELLS (Qidong) Co., Ltd.
Inverter	Type	-	SG250HX
	Number	-	120
	Rated Output power	kW	225
	Rated Output Voltage	V	800
	Rated Output frequency	Hz	50/60
	Manufacturer	-	Sungrow Power Supply Co.,Ltd.

The electricity generation supplied by the project will be sent to the 115/22kV Chrak Mtes Substation by the 22kV transmission line, and then delivered to the EDC. The process flow chart of the project please refers to Figure 1-1.

² 20MWac=26MWp/1.3

³ Sourced from the specifications of the solar cell modules and inverters.

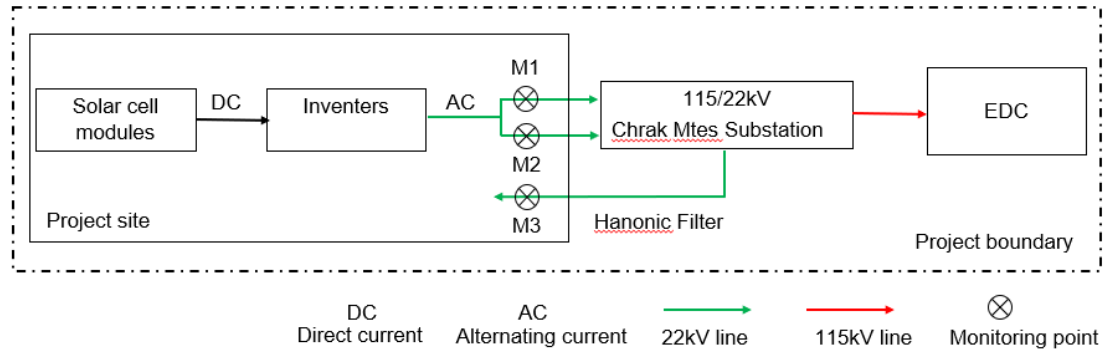


Figure 1-1. The process flow chart of the Project

1.12 Project Location

The project is located in Monourom village, Bavet City, Svay Rieng Province, Cambodia. The geographical coordinates for the project site are east longitude 106°02'49.1" and north latitude 11°05'20.2". The geographic location of the project plant is shown in Figure 1-2.

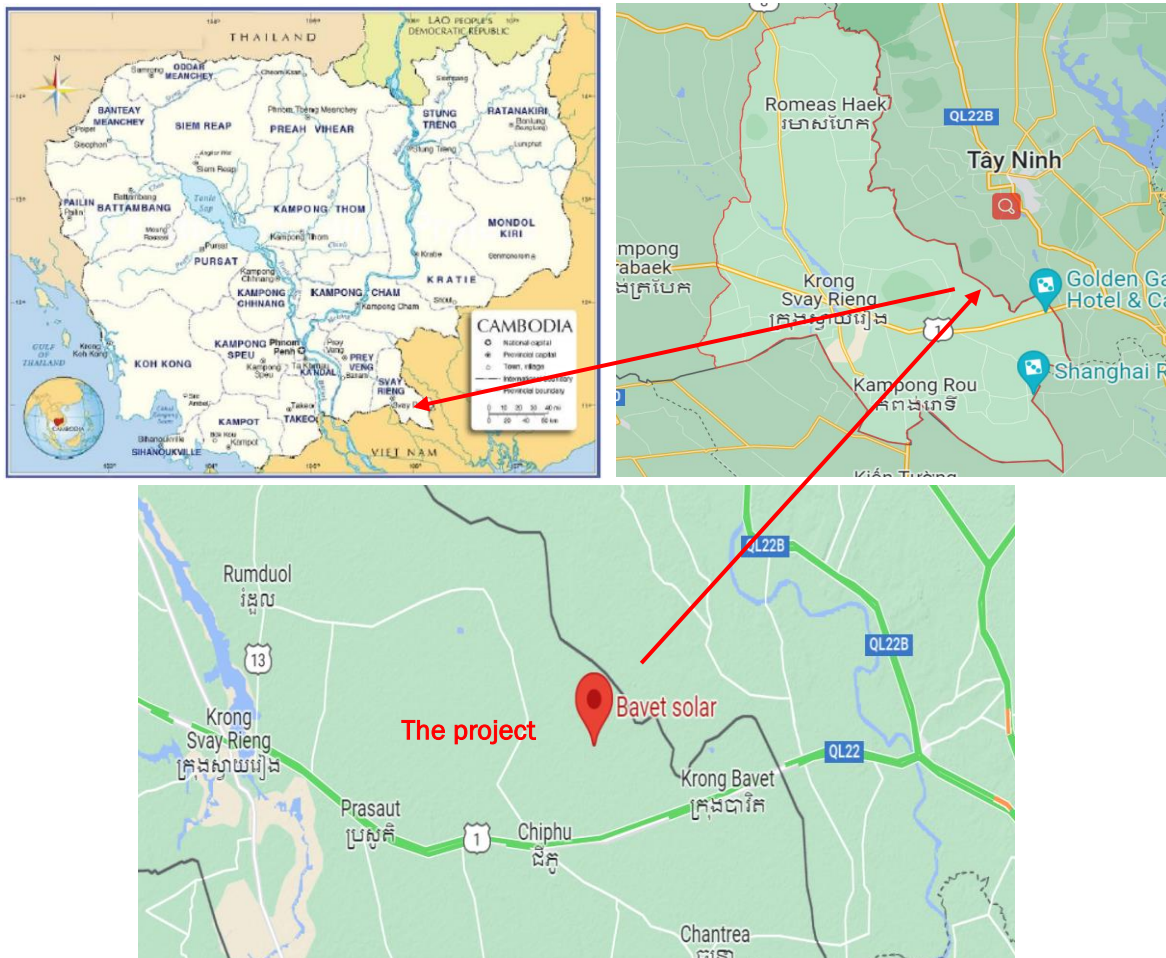


Figure 1-2 The geographic location of the project

1.13 Conditions Prior to Project Initiation

The scenario existing prior to the start of the implementation of the project activity is the electricity delivered to the EDC grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources.

The baseline scenario is the same as the conditions existing prior to the project initiation.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project complies with all Cambodia relevant laws and regulations. Mainly include:

1. The Law on Environmental Protection and Natural Resource Management.
2. Electricity Law of the Kingdom of Cambodia.

The project obtained the project approval and EIA approval from governmental authorities: Ministry of Mines and Energy and Ministry of Environment. The two approvals well demonstrate that local government permits the construction of the project. Consequently, the project is in compliance with laws, status and other regulatory frameworks.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has neither been registered nor seeking registration under any other GHG programs.

The project is seeking registration only in VCS program.

1.15.2 Projects Rejected by Other GHG Programs

Not applicable. The project has not been rejected by any other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes

☒ No

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes

☒ No

1.17 Sustainable Development Contributions

The project is a solar power project with total installed capacity of 26MWp, located in Monourom village, Bavet City, Svay Rieng Province, the Kingdom of Cambodia.



The project will utilize solar resource to generate electricity delivered to the EDC grid which would be generated by the operation of grid-connected power plants and by the addition of new generation sources into the EDC. The project will achieve a GHG emission reduction of 26,473 tCO₂e/yr during the first crediting period. Thus, the project will achieve SDG 13 Climate Action⁴.



The construction and operation of the project will provide 7 employment positions for local villagers, which will have a positive effect on the local economy. During first crediting period, direct and indirect employment opportunities will be generated. Thus, the project will achieve SDG 8 Decent Work and Economic Growth⁵.



38,773 MWh/yr of electricity from renewable sources will be supplied to EDC in the first crediting period. Thus, the project will achieve SDG 7 Affordable and Clean Energy⁶.

1.18 Additional Information Relevant to the Project

Leakage Management

⁴ <https://sdgs.un.org/goals/goal13>

⁵ <https://sdgs.un.org/goals/goal8>

⁶ <https://sdgs.un.org/goals/goal7>

Not applicable as the project is not a AFOLU project.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not applicable as the project does not involve any additional information that may have a bearing on the eligibility of the project, the net GHG emission reductions or removals, or the quantification of the project's net GHG emission reductions or removals.

2 SAFEGUARDS

2.1 No Net Harm

The EIA Report for the project has been approved by Ministry of Environment on 30-Nov-2021. Every aspect of environmental impact has been considered in the EIA report with corresponding measures during project development, and no net harm has been detected. Please refer to section 2.3 for details. Especially for site preparation and civil works, the project owner has planted grass under solar photovoltaic panels to prevent water loss and soil erosion during operation period.



Figure 2-1 The picture of grass under solar photovoltaic panels

Meanwhile, the implementation of the project will improve local socio-economic development through creating career opportunities and paying taxes.

Though the land is sugar plantation prior to construction, no employment on plantations is reported among major income sources, indicating that plantation work is seasonal or an additional livelihood. Additionally, overwhelming majority of the project area households live mainly on farming. A small number of households also run small businesses. A considerable number of people earn their living as workers in garment industry and as construction workers.

In conclusion, the project has no negative impacts on local environment and socio-economy. And no net harm on local environment and social community has been detected for the project.

2.2 Local Stakeholder Consultation

Local Stakeholder Consultation during the project preparation stage:

The super majority of project owner – Pestech was first introduced to the sole owner of project owner back in 2019. In January 2020, Pestech acquired 94% of the equity of the project owner. Salah Essa, a Cambodian individual, owned 6% of the equity of project owner and purchased the project land from previous landowners.

The project was already at the shovel-ready status where the entire land was ready to start together with the 20-year power purchase agreement contract awarded by EDC. The previous land use was entirely a sugar plantation. Pestech did a due diligence exercise to ascertain any unlawful displacement or resettlement of these lands by seeking out and interview the previous owners. All previous landowners had received total land sold fees and there is no economic dispute.



Figure 2-2 The interview with previous landowners

As the entire plot of land was sugar plantation (image appended below) to which the previous owners have agreed to sell, there is no resettlement concern. Furthermore, Cambodia Government has strict laws in place on land acquisition and involuntary settlement in place to provide assurance. Prior to construction, the project has acquired the construction permit issued by Ministry of Land Management on 13-Apr-2021.



Figure 2-3 The landscape of sugar plantation

Prior to the construction, the stakeholder analysis and communication strategy has been prepared by the project owner to serve as a guide for meaningful information disclosure, consultation and active participation of various stakeholders for the project.

As per the stakeholder analysis and communication strategy, the community awareness survey was conducted one month prior to construction by the project owner regarding the scope of the project, procedure of construction activities, utility of resources, the identified impacts and mitigation measures, and grievance redress mechanism. The survey helps the community to resolve problems and clarify their distrusts related to the project at the initial stage.

Based on the results of survey, local stakeholders are supportive of the project construction. The survey shows that a majority of local stakeholders think the project will help improve the life of local people and promote local economic development without much adverse environmental impact. The survey shows that almost all of the stakeholders are supportive to the proposed project, believing that the project will provide more employment opportunities, help local economic development, and increase local power supply.

Therefore, the implementation of the project is regarded as beneficial by most of the local stakeholders.

Local Stakeholder Consultation during the project implementation stage:

Communications with local stakeholders are being carried out at periodic intervals which is every quarter. Key implementation schedules or changes of the project will be communicated to the neighborhood committee and the local residents, the comments and suggestions from residents

will be collected by the project owner and considered during the implementation phase. A grievance signage will be placed out the project gate for anyone to give feedback to.

ជាតិ សាសនា ព្រះមហាក្សត្រ
Nation, Religion, King

គម្រោង: ហ្គ្រីន ស៊ីស្តែនបេស៊ីស វេនឈ័រ (ខេមបូឌា) ឯ.ក
Project: GREEN SUSTAINABLE VENTURES (CAMBODIA) CO., LTD.

គោរពជូនបងប្អូនទាំងអស់គ្នា
ក្នុងករណីអ្នកទាំងអស់មានបញ្ហា ព្រមជាមួយមតិយោបល់ផ្សេងៗ ឬមានការគំរាម ដែលកើតមានឡើង
ដោយសារគម្រោងបេស៊ីសយើងខាងលើ សូមទំនាក់ទំនងតាមព័ត៌មានខាងក្រោមរៀងរាល់ម៉ោងធ្វើការ :

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អ៊ីម៉ែល : info@pestech.com.kh
ម៉ោងធ្វើការ : ច័ន្ទ ដល់ សុក្រ / 8:00ព្រឹក - 5:00 ល្ងាច

Dear Everyone
If you have any grievances, comments or complain cause by our project, please let us know
Below is our contact information (Every working hour)

Phone : (+855) 23 882 105
Email : info@pestech.com.kh
Working Days/Hours : Mon - Fri / 8:00AM - 5:00 PM

In line with VCS requirements all the processes have been implemented to receive comments from local stakeholders as well as communicate with them at periodic intervals.

2.3 Environmental Impact

The final EIA report of the project was approved by Ministry of Environment on 30-Nov-2021. The environmental impacts of the project are summarized as follows.

Topography (Land and vegetation)

As the project site has already been cleared, the site preparation involves a limited amount of vegetation clearance. Site preparation and civil works will change the land use. However, prior to construction, the project has acquired the construction permit issued by Ministry of Land Management on 13-Apr-2021. The procedures of land acquisition are in compliance with the relevant laws in Cambodia.

Vegetation clearances are strictly restricted to the construction sites temporary areas and access roads. No chemicals (herbicides) are used. Loose soil from foundations is restored through ramming. Excess soil is spread out in areas such that it does not disturb the local drainage pattern. Engineering and biological measures are implemented to prevent soil erosion such as sow soil binding grass. Landscaping/tree plantation with native species along the project site

fenced area will be done as soon as earthworks are completed to stabilize the soil and reduce dust pollution.

During the operation period, periodic vegetation maintenance (e.g. trimming of shrubs, grass and tree branches) will be undertaken at the project site and within the transmission right of way in order to maintain necessary safety clearances. Manual vegetation maintenance methods will be used, and local labor will be utilized where possible creating a potential job opportunity.

Air pollution

In the construction period, opened and exposed land areas at the project site, drilling activities, loose dust at the base of the panel structures, are sprayed with water to suppress dust levels particularly the dry season. Aside from this, the construction site is temporarily enclosed to control dust level. Construction vehicles transporting materials that generate dusts are covered with tarps. During the operation period, the project does not involve exhaust gas emissions.

Water pollution

In the construction period, the EPC contractor provided workers with sanitary facilities to treat domestic wastewater or sewage and complied with National Sub-decree No.27 on Water Pollution Control.

The wastewater generated during the operation period is solar panels cleaning wastewater. The solar panels cleaning wastewater mainly includes dust, which does not require chemical treatment. Domestic wastewater can be recycled and reused for gardening. Sewage is directed to onsite system associated with sanification facilities.

Noise

In the construction period, the use of equipment, construction vehicles and civil works may increase the noise levels such as during excavation works at the project site and induce vibration, which impact buffer area on temporary basis.

During the operation period, the main noise is generated by inverters and transformers. Operation of noise-generating equipment are enclosed, and periodic maintenance of equipment will be conducted. Noise level measurements has been conducted to ensure that noise level is complied with relevant national standard. After distance attenuation, there is basically no noise contribution to the surrounding boundary during the project production, and it has little influence on the boundary acoustic environment.

Solid waste

A covered sorting area for recyclables is established on-site in the construction period. Wastes such as packaging materials, meta and wires, electronics, wood (if any), are transported to EDC depots/dedicated storage yards, for resale and auction to authorized dealers. Multiple waste containers are also be provided. Waste burning in and around the site including buffer area are prohibited.

In the operation period, the solid waste generated after the decommissioning is waste equipment and waste solar panels, licensed companies can be hired to collect, transport, and dispose of unused hazardous materials and wastes.

In conclusion, the environmental impact during the project construction and operation period will be minor. The Project owner has taken appropriate measures to minimize adverse environmental impacts.

2.4 Public Comments

This project will be open for public comment on the verba website.

2.5 AFOLU-Specific Safeguards

Not applicable as the project is non-AFOLU.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Applied approved CDM methodology: ACM0002 Grid-connected electricity generation from renewable sources (Version 21.0)

This methodology also refers to the latest approved version of the following tools:

“Tool to calculate the emission factor for an electricity system” (version 07.0)

“Tool for the demonstration and assessment of additionality” (version 07.0.0)

“Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (version 3.0)

“Common practice” (version 03.1)

“Investment analysis” (version 11.0)

For more detail information about the methodology and tools, please reference to the following link: <https://cdm.unfccc.int/methodologies/DB/XP2LKUSA61DKUQCOPWPGWDN8ED5PG>

3.2 Applicability of Methodology

The project satisfies all the applicability criteria of the methodology ACM0002 (Version 21.0), of which the detailed description is listed in Table 3-1 below:

Table 3-1 Applicability of ACM0002

Para No.	Applicability criteria	Justification
1	This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant. (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plants/units. (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s).	Applicable. The project is a greenfield solar PV power plant. The project belongs to condition (a).
2	In case the project activity involves the integration of a Battery Energy Storage System (BESS), the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s).	Not applicable. The project does not involve the integration of a BESS.
3	The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity; (c) In case of Greenfield project activities applicable under paragraph 5 (a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g. by referring to feasibility studies or investment decision documents);	Applicable. The project is a greenfield solar PV power plant. The project activity does not involve capacity additions, retrofits, rehabilitations or replacement, and the integration of a BESS. The project belongs to condition (a).

	(d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g. week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period.	
4	In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply. (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity. (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be a. Lower than or equal to 15 MW; and	This condition is not relevant, as the project activity is not the installation of a hydro power plant.

	b. Less than 10 per cent of the total installed capacity of integrated hydro power project	
5	In the case of integrated hydro power projects, project proponent shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	This condition is not relevant, as the project activity is not the installation of a hydro power plant.
6	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units;	The project activity does not involve any of the given criteria hence methodology is applicable for the project activity.
7	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	This condition is not relevant, as the project activity does not involve capacity additions, retrofits, replacement or rehabilitations.

In addition, the project meets the applicability conditions of the applied tools applied in the PD as follows:

Table 3-2 Applicability of applied tools

Tool	Applicability criteria	Justification
Tool for the demonstration and	The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for	Applicable

assessment of additionality (Version 07.0.0)	project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	The methodology selected for the proposed project requires the use of this tool.
	Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	Applicable The methodology applied in this proposed project requires the use of this tool.
Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03.0)	If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer. (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity	Applicable The electricity consumption by the Baseline, project and/or leakage emissions is provided by the EDC, which is Scenario A.

	consumer can be provided with electricity from the captive power plant(s) and the grid.	
	This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid. (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.	Applicable The electricity generated in the project scenario is supplied to the EDC, which is Scenario I.
	This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO₂ emissions.	Applicable No captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage.
Tool to calculate the emission factor for an electricity system (Version 07.0)	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	Applicable This project replaces grid power supply and uses this tool to calculate the values of OM, BM and CM of this project.
	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants	Based on the public data in Cambodia, either the specific fuel consumption of the EDC grid system in most recent five years or CO₂ emission factor of each plant are not available, lead to the combined margin emission factor ($EF_{grid,CM,y}$) cannot be calculated by this

	(in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	tool. Besides, there is no available latest emission factor of the EDC grid as approved by the relevant National Authority or Designated National Authority (DNA) in Cambodia. The average grid emission factor $EF_{grid,CM,y}$ of 0.6828 CO₂/MWh which is calculated based on three reliable data source is applied directly to calculate emission reduction. Please refer to section 3.5 for details.
	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	This condition is not relevant, there is no part of the power system of this project located in Annex I countries.
	Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	This condition is not relevant, this project is a solar power project.
Investment analysis (Version 11.0)	This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario	Applicable The project applies the methodological tool “Tool for the demonstration and assessment of additionality”.
	In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the	The methodology ACM0002 (Version 21.0) applied in this project requires the use of this tool to demonstrate the

	requirements contained in the methodology shall prevail.	investment analysis of this project.
Common practice (Version 03.1)	This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality.	Applicable This project applies the methodological tool “Tool for the demonstration and assessment of additionality”.
	In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The methodology ACM0002 (Version 21.0) applied in this project requires the use of this tool to demonstrate the common practice of this project.

3.3 Project Boundary

According to ACM0002 (Version 21.0), the spatial extent of the project boundary includes the project power plant/unit and all power plants/units connected physically to the EDC grid that the project power plant is connected to. The project boundary is shown in Figure 3-1 below.

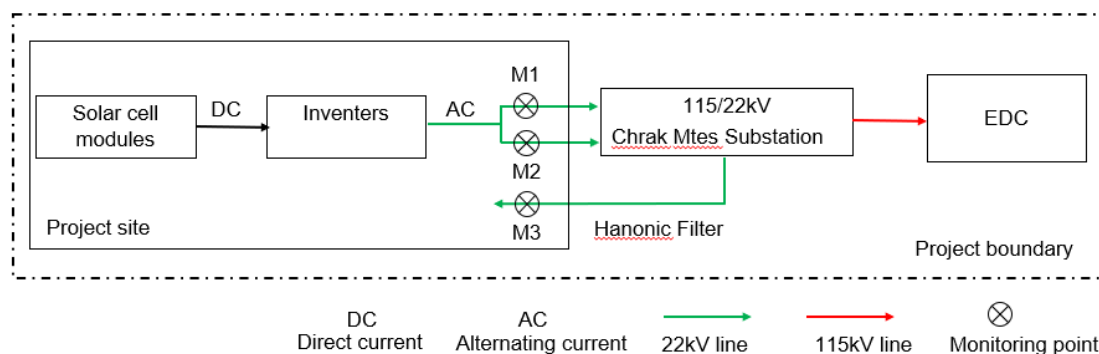


Figure 3-1 Diagram of the project boundary

The main emission sources and gases included in the project boundary is listed in table below:

Source	Gas	Included?	Justification/Explanation
Baseline CO ₂ emissions from electricity	CO ₂	Yes	The major source of emissions in the baseline.

Source		Gas	Included?	Justification/Explanation
	generation in fossil fuel fired power plants that are displaced due to the project activity	CH ₄	No	Excluded for simplification. This emission source is assumed to be very small.
		N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small
Project	Emission Sources of Project Activity	CO ₂	No	According to the ACM0002, the production and operation of solar photovoltaic projects do not produce significant greenhouse gas emissions. Thus, the project emissions can be ignored.
		CH ₄	No	
		N ₂ O	No	

3.4 Baseline Scenario

According to the methodology ACM0002 (Version 21.0), as the project is the installation of a Greenfield solar power plant, the baseline scenario of the project is electricity delivered to EDC by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources.

3.5 Additionality

According to the methodology ACM0002 (Version 21.0), the additionality of the project activity shall be demonstrated and assessed using the latest version of the “TOOL01: Tool for the demonstration and assessment of additionality”. The steps are listed as following.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Define realistic and credible alternatives to the project activity(s) through the following Sub-steps.

Sub-step 1a: Define alternatives to the project activity

Alternatives available to the project participant or similar project developers that provide outputs or services comparable with the project activity are identified below:

Alternative (a) The proposed project activity undertaken without being registered as a VCS project activity.

Alternative (b) Continuation of the current situation (no project activity or other alternatives undertaken).

Sub-step 1b: Consistency with mandatory laws and regulations

Both alternatives (a) and (b) are realistic and credible alternatives to the project which are consistent with mandatory laws and regulations.

Step 2: Investment analysis

Sub-step 2a: Determine appropriate analysis method

According to ACM0002 (Version 21.0), if the project activity and the alternatives identified in Step 1 generate no financial or economic benefits other than carbon related income, then apply the simple cost analysis (Option I). Otherwise, use the investment comparison analysis (Option II) or the benchmark analysis (Option III).

For the project, the alternative to the project activity is the supply of electricity from a grid which is not considered as an investment by project participant, so the benchmark analysis (Option III) is used.

Sub-step 2b: Option III. Apply benchmark analysis

According to “Investment analysis” (version 11.0), “Local commercial lending rates or weighted average costs of capital (WACC) are appropriate benchmarks for a project IRR”. The local lending rate offered by National Bank of Cambodia (NBC) is chosen by the project entity for investment analysis. Since the project owner sent the letter for the project to Ministry of Mines and Energy (MME) on 10-May-2019 and approved by MME on 12-Jul-2019, Cambodian average commercial bank prime lending rate 2018 (10.61%⁷) is chosen as the benchmark for the project investment analysis.

Based on above benchmark, calculation and comparison of financial indicators are carried out in sub-step 2c.

Sub-step 2c: Calculation and comparison of financial indicators

Due to the impact of loan interest on income tax calculations, it is recommended that when a project IRR is calculated to demonstrate additionality a pre-tax benchmark be applied. Therefore, pre-tax project Internal Rate of Return (IRR) is calculated in investment analysis. The pre-tax project IRR is compared to the benchmark of 10.61% defined in sub-step 2b.

The main financial parameters and IRR of the project are shown in Table 3-3 below.

Table 3-3 Main parameters for financial analysis calculation

Item	Unit	Value	Source
Installed capacity	MWp	26	Financial investment analysis sheet prior to the construction of the
Total static investment (excluding VAT or other	USD	24,718,000	

⁷ NBC Annual Report 2018 issued on 30-Jan-2019, <https://www.nbc.org.kh/english/index.php>

taxes in Cambodia)			project
Average annual electricity exported	MWh/year	37,085	
Electricity Tariff (excluding VAT)	USD/MWh	76	
Depreciation period	Year	20	
Depreciation rate	%	5	
Average annual operation and management cost (O&M)	USD/year	398,972	
Construction period	Year	1	
Operation period	Year	20	
Expected VCU Price	USD/tCO ₂ e	7	

Table 3-4 below shows the difference between the project IRR, with and without the revenue from carbon.

Table 3-4 IRR results with and without carbon credits

Item	Without carbon revenue	Benchmark rate	With carbon revenue
Project IRR (before tax)	7.88%	10.61%	9.14%

Sub-step 2d: Sensitivity analysis

The sensitivity analysis is to show whether the conclusion regarding the financial/economic attractiveness is robust to reasonable variations in the critical assumptions.

According to “Investment analysis” (version 11.0), only variables, including the initial investment cost, that constitute more than 20% of either total project costs or total project revenues should be subjected to reasonable variation. For the project, the total static investment constitutes more than 20% of total project costs; the product of electricity tariff and Annual electricity delivered to the grid constitute more than 20% of the total revenue of the project. And the total O&M throughout the project lifetime accounts for more than 20% of the project cost.

Therefore, considering the reasonable variations in the critical assumptions, the sensitivity analysis is made regarding the 4 key factors of total static investment, electricity tariff, annual O&M cost and annual electricity delivered to the grid, as shown in Table 3-5.

Table 3-5 Sensitivity analysis of IRR for the project.

Hypothetic change	-10%	-5%	0	+5%	+10%
Total Static investment	9.49%	8.66%	7.88%	7.17%	6.51%
Annual electricity delivered to the grid	6.26%	7.08%	7.88%	8.67%	9.44%
Annual O&M cost	8.10%	7.99%	7.88%	7.78%	7.67%
Electricity tariff	6.26%	7.08%	7.88%	8.67%	9.44%

Table 3-6 below shows the necessary change in the four major parameters required for the Project IRR without VCU's revenue to reach the benchmark rate, ceteris paribus. All the speculated changes are clearly too large to be realistic.

Table 3-6 Critical Point Analysis

Hypothetic Project IRR	Total static investment	Annual electricity delivered to the grid	Annual O&M cost	Electricity tariff
Benchmark Rate 10.61%	-16.00%	+17.83%	-132.48%	+17.83%

Total static Investment

The Project IRR would reach the benchmark rate of 10.61% when the total static investment decreases by 16.00%. According to the investment details by the project owner, the actual investment of the project is almost same as the total static investment. Therefore, it is impossible that the total static investment can be reduced by 16.00% to make the IRR to cross the benchmark.

Annual electricity delivered to the grid

The Project IRR would reach the benchmark rate of 10.61% when annual electricity delivered to the grid increased by 17.83%. Annual electricity delivered to the grid is determined by the average annual power generation which is linked directly to the local solar resource. The projected power generation is estimated based on long term weather statistic data and the annual degradation rate of solar PV modules. Therefore, an increase of 17.83% for annual electricity delivered to the grid is basically impossible.

Annual O&M cost

Variations in the annual O&M costs have an insignificant impact on the return of the project. The IRR cannot cross benchmark when the O&M cost is zero, and it is impossible to happen at all.

Electricity tariff

The Project IRR would reach the benchmark rate of 10.61% when annual electricity tariff increased by 17.83%. According to electricity purchase contract of the project and sales receipts, electricity tariff is 76 USD/MWh which is the same as the one used in financial investment analysis. Therefore, it is not like for the project to get 17.83% increase of electricity tariff to cross the benchmark.

In conclusion, the investment analysis and sensitivity analysis show that the project undertaken without carbon credit revenue is not financially attractive and the carbon revenue will improve the financial indicators of the Project.

Step 3: Barrier analysis

The proposed project does not employ the barrier analysis. Then proceed to Step 4.

Step 4: Common practice analysis

As the project applies power generation based on renewable energy which is one of the measure(s) listed in the definitions section above, proceed to Sub-step 4a.

Sub-step 4a: The proposed project activity(ies) applies measure(s) that are listed in the definitions section above

Common practice analysis is carried out as per “Common practice” (version 03.1).

Sub-Step 4a-1: calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity

The install capacity of the project is 26 MWp, so the applicable capacity range as +/-50% of the total design capacity is 13 ~ 39 MWp.

Sub-Step 4a-2: identify similar projects which fulfil all of the following conditions:

Cambodia is selected as applicable geographical area. As the starting date of the proposed project is 13-Jan-2020 which is the date of signing EPC contract, so the solar power plants of the capacity from 13 MWp to 39 MWp operational before 13-Jan-2020 are identified.

There are no solar power plants with capacity from 13 MWp to 39 MWp operational in Cambodia before 13-Jan-2020 based on available published sources⁸.

Sub-step 4a-3: *Within the projects identified in Sub-step 4a-2, identify those that are neither registered CDM/VCS/CCER project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number N_{all} .*

⁸ https://www.edc.com.kh/Annually_page/annuallyReport

<https://eac.gov.kh/>

According to **Sub-Step 4a-2**, there are no solar power plants with capacity from 13 MWp to 39 MWp operational in Cambodia before 13-Jan-2020, $N_{all} = 0$.

Sub-step 4-4: *Within similar projects identified in Sub-step 4-3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number N_{diff} .*

For the project, $N_{diff} = 0$.

Sub-step 4-5: *Calculate factor $F = 1 - N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.*

Since $N_{all} = 0$, $N_{diff} = 0$, therefore, $N_{all} - N_{diff} = 0$, which is less than 0.

$F = 1 - N_{diff}/N_{all} = 1$.

According to common practice (version 03.1), the proposed project activity is a “common practice” within a sector in the applicable geographical area if the factor F is greater than 0.2 and $N_{all} - N_{diff}$ is greater than 3. However, the $N_{all} - N_{diff}$ is less than 3.

Thus, the Project activity is not a “common practice”.

Outcome of Step 4:

Based on the analysis above, the baseline scenario of the project is:

Electricity delivered to EDC by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources.

Therefore, the proposed project is additional.

3.6 Methodology Deviations

Based on the public data in Cambodia, either the specific fuel consumption of the EDC grid system in most recent five years or CO₂ emission factor of each plant are not available, lead to the combined margin emission factor ($EF_{grid,CM,y}$) cannot be calculated by the latest version “Tool to calculate the emission factor for an electricity system”. Besides, there is no available latest emission factor of the EDC grid as approved by the relevant National Authority or Designated National Authority (DNA) in Cambodia.

By checking the UNFCCC website, for the purpose of promoting greater harmonization, the IFI Technical Working Group (IFI TWG) on GHG accounting maintains a common dataset containing Default Emissions Factor (DEF) of the country’s electricity grid including in-country interconnected grids. The DEFs apply to electricity generation in a country and currently do not consider the impact of interconnections with neighbouring countries. The common dataset containing DEFs is constructed using a Combined Margin (CM) for the grid that is comprised of an Operating Margin (OM) and a Build Margin (BM). According to the latest IFI Dataset of Default

Grid Factors v.3.2 published in April 2022⁹, the latest $EF_{grid,CM,y}$ of solar project in Cambodia in 2021 is 0.8743 tCO₂e/MWh which is authoritative and closer to the operation start date of the project. Thus, 0.8743 tCO₂e/MWh can be used to calculate emission reduction for the project.

Additionally, as per IGES List of Grid Emission Factors version 10.12¹⁰, 2009 Phnom Penh grid emission factors of solar project in Cambodia is 0.6413 tCO₂e/MWh which is conservative but not the latest data. By checking the JCM website, in case the PV system in a proposed project activity in Cambodia is connected to an internal grid which is not connected to the national grid, the emission factor is set to 0.533 tCO₂/MWh from JCM Approved Methodology KH_AM002¹¹.

To be conservative and reasonable, the average grid emission factor $EF_{grid,CM,y}$ was calculated as 0.6828¹² tCO₂/MWh based on above three data source.

The deviation has no impact on conservativeness of the quantification of GHG emission reductions or removals. The deviation relates only to the criteria and procedures for monitoring or measurement and does not relate to any other part of the methodology.

4 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The GHG emission reduction calculation of the project was based on the applied methodology ACM0002 (Version 21.0).

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y} \quad (1)$$

Where:

⁹ <https://unfccc.int/climate-action/sectoral-engagement/ifi-harmonization-of-standards-for-ghg-accounting/ifi-twg-list-of-methodologies>

¹⁰ <https://www.iges.or.jp/en/pub/list-grid-emission-factor/en>

¹¹ https://www.jcm.go.jp/kh-jp/methodologies/46/approved_pdf_file

¹² 0.6828 tCO₂/MWh = (0.8743 tCO₂/MWh + 0.6413 tCO₂/MWh + 0.533 tCO₂/MWh) / 3

BE_y	=	Baseline emission in year y (tCO ₂ e)
$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM-project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	=	Combined margin emission factor (tCO ₂ e/MWh)

The project is a Greenfield solar power plant, then:

$$EG_{PJ,y} = EG_{facility,y} \quad (2)$$

Where:

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr).

Following ACM0002 (Version 21.0), the baseline emission factor should be calculated as combined margin emission factor of EDC ($EF_{grid,CM,y}$), consisting of the combination of operating margin emission factor of EDC ($EF_{grid,OM,y}$) and build margin emission factor of EDC ($EF_{grid,BM,y}$) according to the following six steps defined in the “Tool to calculate the emission factor for an electricity system” (version 7.0).

However, based on the public data in Cambodia, either the specific fuel consumption of the EDC grid system in most recent five years or CO₂ emission factor of each plant are not available, lead to the combined margin emission factor ($EF_{grid,CM,y}$) cannot be calculated by the latest version “Tool to calculate the emission factor for an electricity system”. Besides, there is no available latest emission factor of the EDC grid as approved by the relevant National Authority or Designated National Authority (DNA) in Cambodia.

Thus, according to the latest IFI Dataset of Default Grid Factors v.3.2 published in April 2022¹³, IGES List of Grid Emission Factors version 10.12¹⁴ and JCM Approved Methodology KH_AM002¹⁵, the calculated average $EF_{grid,CM,y}$ of 0.6828 tCO₂e/MWh for solar project in Cambodia are applied in the project and fixed in the first crediting period.

As per the PVsyst report, the annual electricity supplied to the grid is shown in the table below. The $EF_{grid,CM,y}$ of the project is 0.6828 tCO₂e/MWh, therefore baseline emission of the project is calculated as follows:

¹³ <https://unfccc.int/climate-action/sectoral-engagement/ifis-harmonization-of-standards-for-ghg-accounting/ifi-twg-list-of-methodologies>

¹⁴ <https://www.iges.or.jp/en/pub/list-grid-emission-factor/en>

¹⁵ https://www.jcm.go.jp/kh-jp/methodologies/46/approved_pdf_file

Year	EG _{facility,y} (MWh)	EF _{grid,CM,y} (t CO _{2e} /MWh)	BE _y (t CO _{2e})
	A	B	C=A*B
27-Jan-2022~31-Dec-2022	36,555	0.6828	24,959
01-Jan-2023~31-Dec-2023	39,176	0.6828	26,749
01-Jan-2024~31-Dec-2024	38,980	0.6828	26,615
01-Jan-2025~31-Dec-2025	38,785	0.6828	26,482
01-Jan-2026~31-Dec-2026	38,591	0.6828	26,349
01-Jan-2027~31-Dec-2027	38,398	0.6828	26,217
01-Jan-2028~31-Dec-2028	38,206	0.6828	26,086
01-Jan-2029~26-Jan-2029	2,721	0.6828	1,857
Total	271,411		185,314

In conclusion, the total estimated baseline emission of the project in the first crediting period is 185,314 tCO_{2e}.

4.2 Project Emissions

As a solar power generation project, according to ACM0002 (Version 21.0), the project emissions $PE_y = 0$.

4.3 Leakage

According to ACM0002 (Version 21.0), no leakage emissions are considered.

4.4 Estimated Net GHG Emission Reductions and Removals

Net GHG Emission Reductions are calculated as follows:

$$ER_y = BE_y - LE_y - PE_y \quad (3)$$

Where

ER_y = Emission reductions in year y (tCO_{2e})

BE_y = Baseline emissions in year y (tCO_{2e})

PE_y = Project emissions in year y (tCO_{2e})

LE_y = Leakage emissions in year y (tCO_{2e})

Since $LE_y = 0$ and $PE_y = 0$

$$ER_y = BE_y$$

In first crediting period, the total estimated emission reduction of the project is 185,314 tCO_{2e}, with average annual emission reductions of 26,473 tCO_{2e}.

The annual emission reductions during first crediting period are listed in below table.

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
27-Jan-2022~31-Dec-2022	24,959	0	0	24,959
01-Jan-2023~31-Dec-2023	26,749	0	0	26,749
01-Jan-2024~31-Dec-2024	26,615	0	0	26,615
01-Jan-2025~31-Dec-2025	26,482	0	0	26,482
01-Jan-2026~31-Dec-2026	26,349	0	0	26,349
01-Jan-2027~31-Dec-2027	26,217	0	0	26,217
01-Jan-2028~31-Dec-2028	26,086	0	0	26,086
01-Jan-2029~26-Jan-2029	1,857	0	0	1,857
Total	185,314	0	0	185,314

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ /MWh
Description	Combined margin emission factor
Source of data	The IFI Dataset of Default Grid Factors version 3.2, IGES List of Grid Emission Factors version 10.12 and JCM Approved Methodology KH_AM002
Value applied:	0.6828

Justification of choice of data or description of measurement methods and procedures applied	The $EF_{grid,CM,y}$ of the project is calculated based on the grid emission factors from three different data source.
Purpose of Data	Calculation of baseline emissions
Comments	Fixed in the first crediting period

5.2 Data and Parameters Monitored

Data / Parameter	$EG_{facility,y}$
Data unit	MWh/yr
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Source of data	Electricity meter
Description of measurement methods and procedures applied	Monitored by electricity meter(s) and monthly recorded by the project operator.
Frequency of monitoring/recording	Monitoring continuously and recording monthly
Value applied:	The ex-ante value please refer to section 4.1, and the parameter will be monitored in the monitoring period.
Monitoring equipment	Electricity meters
QA/QC procedures applied	The data will be cross checked with sale and purchase receipts. The electricity meter(s) will be periodically checked and maintained. Calibration will be carried out periodically to ensure normal operation. All electronic data and paper documents will be archived for two years following the end of the crediting period.
Purpose of data	Calculation of baseline emissions

Calculation method	Calculated as difference between (a) the quantity of electricity supplied by the project plant to the grid and (b) the quantity of electricity delivered to the project plant from the grid.
Comments	-

5.3 Monitoring Plan

The monitoring plan presented in this report assures that real, measurable, long-term GHG emission reductions can be monitored, recorded and reported. It is a crucial procedure to identify the final emission reduction of the project. This monitoring plan will be implemented by the project owner during the project operation. The details of the monitoring plan are specified as follows:

Management Structure

The Project owner organizes a specific monitoring team to be responsible for data collection, supervision and witness the whole process of data measuring and recording. Monitoring manager is appointed to take full responsibility for the overall monitoring of the project. The monitoring, recording of electricity meters is to be carried out by designated monitoring staff. In addition, the Project developer appoints internal verifiers who is responsible for internal check of the measurement, collection of relevant sales and purchase receipts, and the calculation of the emission reductions.

Data and parameters to be monitored

The parameter to be monitored is Quantity of net electricity generation supplied by the project plant/unit to the grid, which is calculated as difference between (a) the quantity of electricity supplied by the project plant to the grid and (b) the quantity of electricity delivered to the project plant from the grid.

Both (a) and (b) are to be continuously measured by electricity meters installed at the 22kV line in the high voltage side of the project. Meter readings will be at least monthly recorded. The accuracy of all electricity meters is not lower than 0.5s, and the meters will be calculated regularly.

Data collection

Monitoring staffs of the proposed project are responsible for data collection. Designated teams will read and collect the monitored data regularly and recorded in the meter reading records or operation log. The meter readings are also available to the power grid through remote reading system. Monthly settlement for electricity supplied to the grid will be based on the meter readings. Meter reading records will serve as the main data source for emission reductions calculation, after crosschecking with the sales and purchase receipts. All data files, relevant sales receipts will be collected by designated monitoring staff, who will prepare backup in time and archive all documents properly.

Quality assurance

The electricity meter installed shall follow the requirements of national industry standard to ensure sufficient accuracy and proper calibration. The electricity meter of the project is installed at the 22kV line in the high voltage side of the project, the accuracy of electricity meters is not lower than 0.5s, and the calibration shall be carried out regularly.

Error check routines will be established on site and at the point of data storage to detect data measuring/transmission failures as well as malfunctions. In the case of malfunction of the meters, the meter supplier will provide technical support to engage the problem promptly and emission reductions during the corresponding period will be calculated conservatively.

Data file management

All monitoring data will be electronically filed by the end of each month and the electronic data files will be archived in both disk copy and printed hard copy. Other documents in paper e.g., operation logbooks, will be preserved as well. All data collected as part of monitoring will be archived electronically and be kept at least for 2 years after the end of the crediting period.